

Are You Smarter than DoRA?

Ryan Ye, Boaz Ng, Leon Jiao, Aadi Singla

Dora's Motivation

Can she solve interesting math problems using efficient fine-tuning?

Methods

LoRA: Low Rank Weight Update

$$W' = W_0 + BA$$

Key DoRA Insight: Separate Magnitude and Direction

$$W' = \frac{W_0 + BA}{\|W_0 + BA\|_c}$$

0.71% of total trainable params!
0% inference latency!

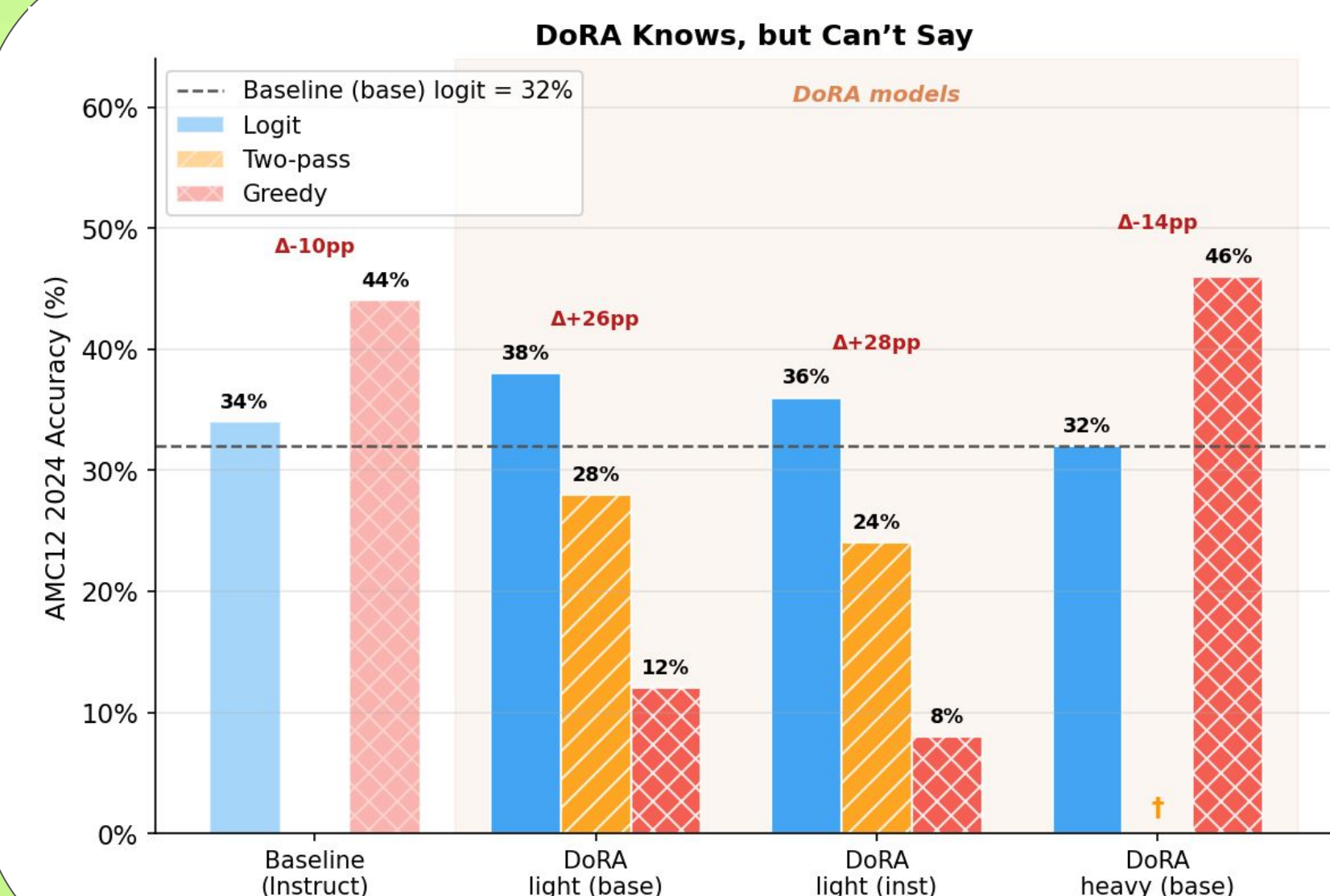
Model: Qwen 2.5-3B; Data: MATH-lighteval NuminaMath-CoT; Eval: AMC 12, 24/25

Problem

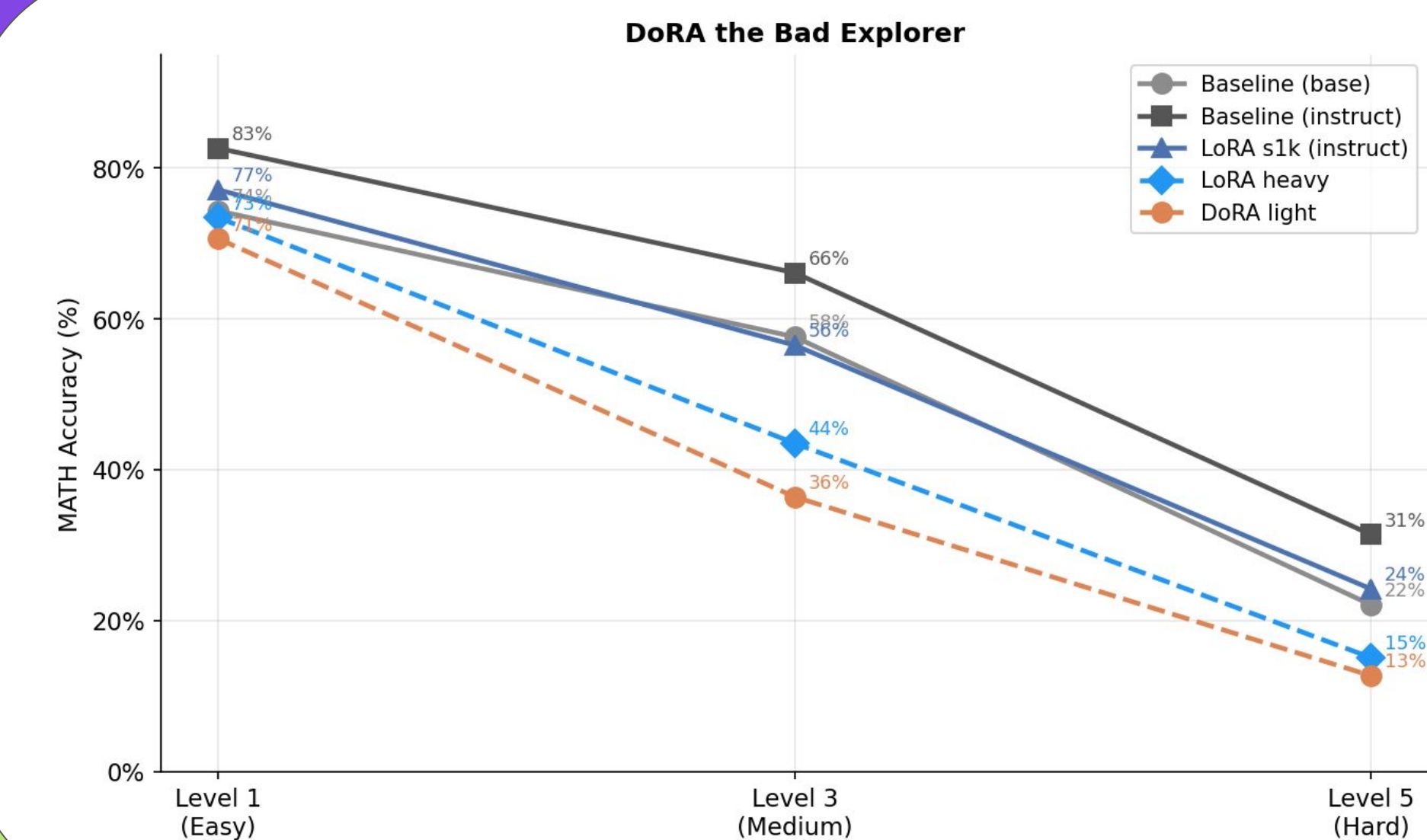
The product of three integers is 60. What is the least possible positive sum of the three integers?

(A) 2 (B) 3 (C) 5 (D) 6 (E) 13

Results



Conclusion



Dora's Next Mission

- Harder tasks – AIME / Math Olympiad
- Alternative decoding interventions
- Bigger base model

Citations

- [1] Liu, S.-Y., et al. *DoRA: Weight-Decomposed Low-Rank Adaptation*. ICML 2024.
- [2] Hu, E. J., et al. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR 2022.
- [3] DigitalLearningGmbH. *MATH-lighteval dataset*, Hugging Face, 2024.
- [4] Li, J., et al. *NuminaMath-CoT dataset*, AI-MO, Hugging Face, 2024.
- [5] Mangrulkar, S., et al. *PEFT: State-of-the-art Parameter-Efficient Fine-Tuning*. Hugging Face, 2022.



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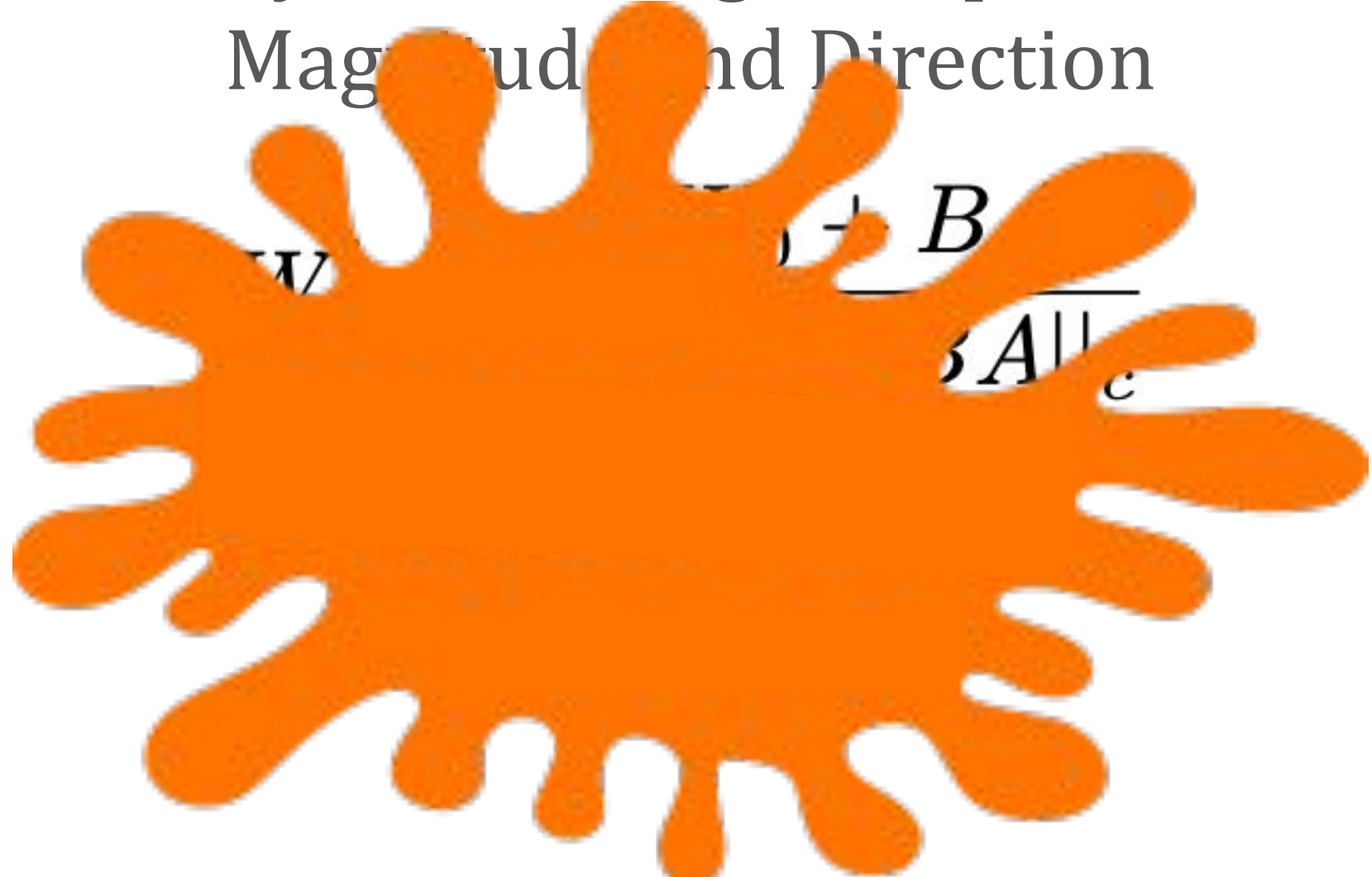
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Methods

LoRA: Low Rank Weight Update

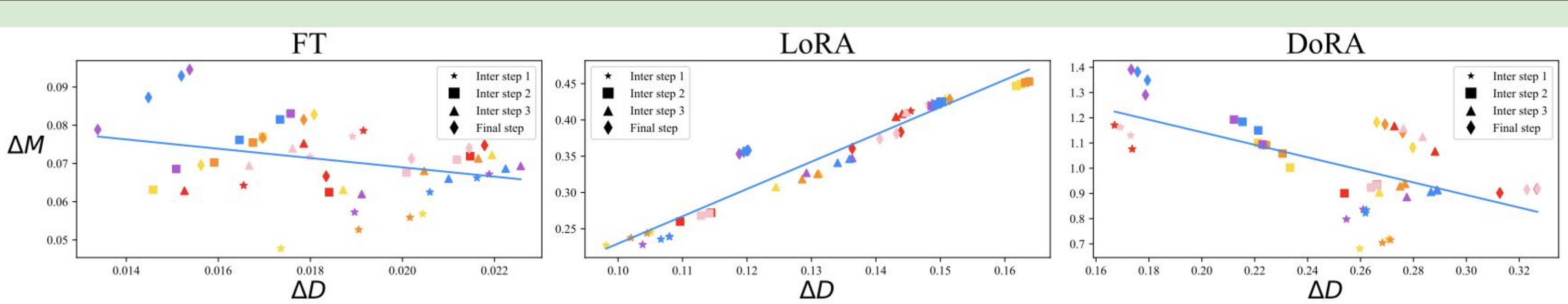
$$W' = W_0 + BA$$

Key DoRA Insight: Separate Magnitude and Direction



0.71% of total trainable params!!!!
0% inference latency!!!!

Q_proj → Q_proj with DoRA update



Can we solve difficult math problems using efficient fine-tuning?

(Insert results)
DoRA vs LoRA vs Base Model

Problem

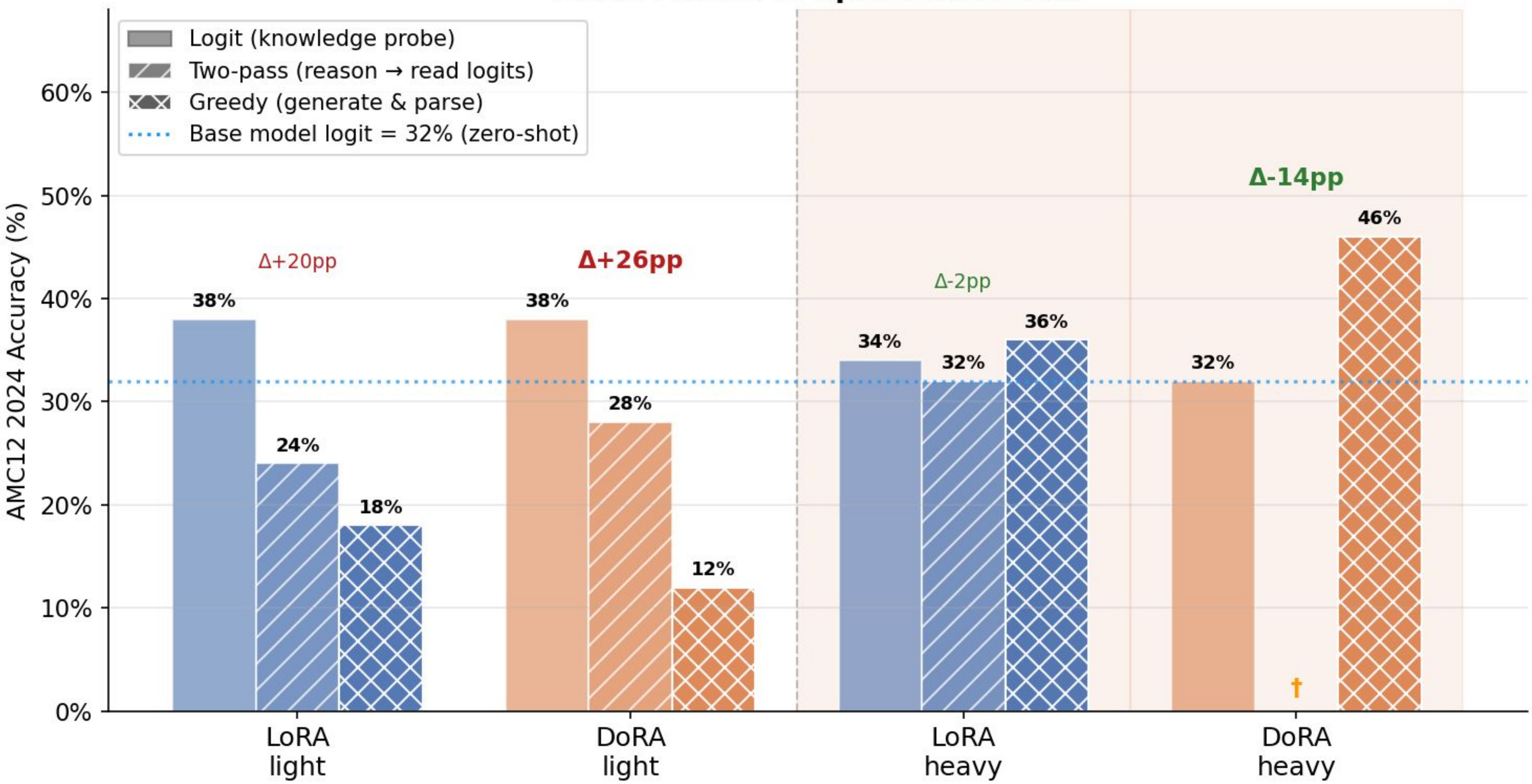
Carlos uses a 4-digit passcode to unlock his computer. In his passcode, exactly one digit is even, exactly one (possibly different) digit is prime, and no digit is 0. How many 4-digit passcodes satisfy these conditions?

- (A) 176 (B) 192 (C) 432 (D) 464 (E) 608

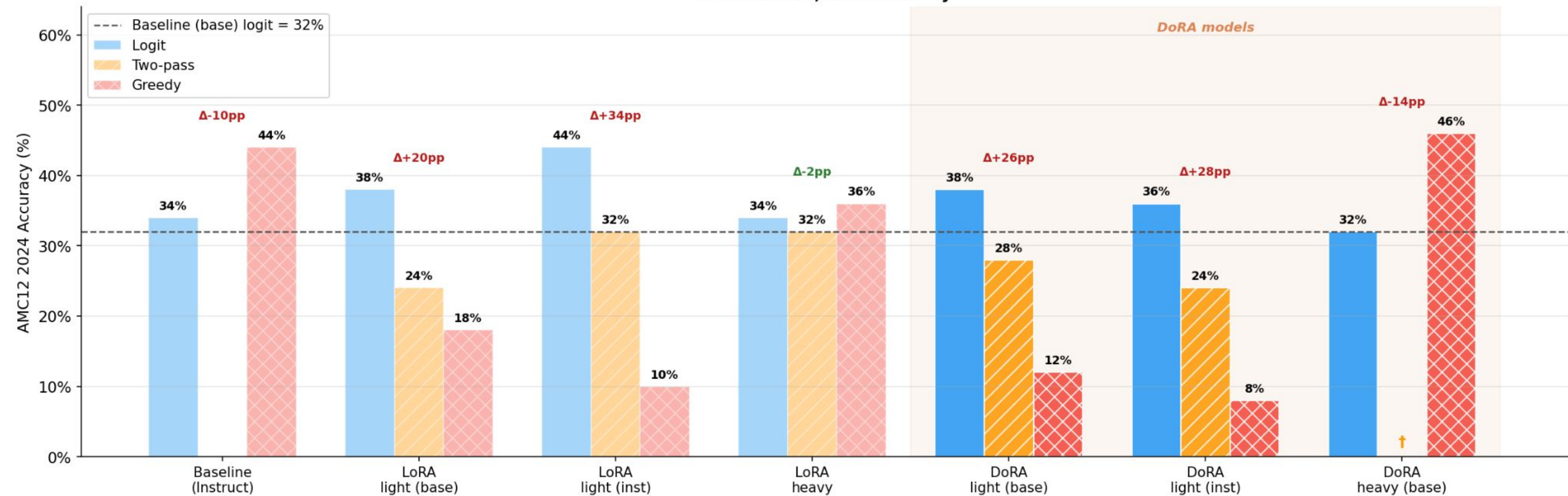
Model: Qwen 2.5-3B
Data: MATH-lighteval,
NuminaMath-CoT
Eval: AIME/AMC12 24/25

Citations

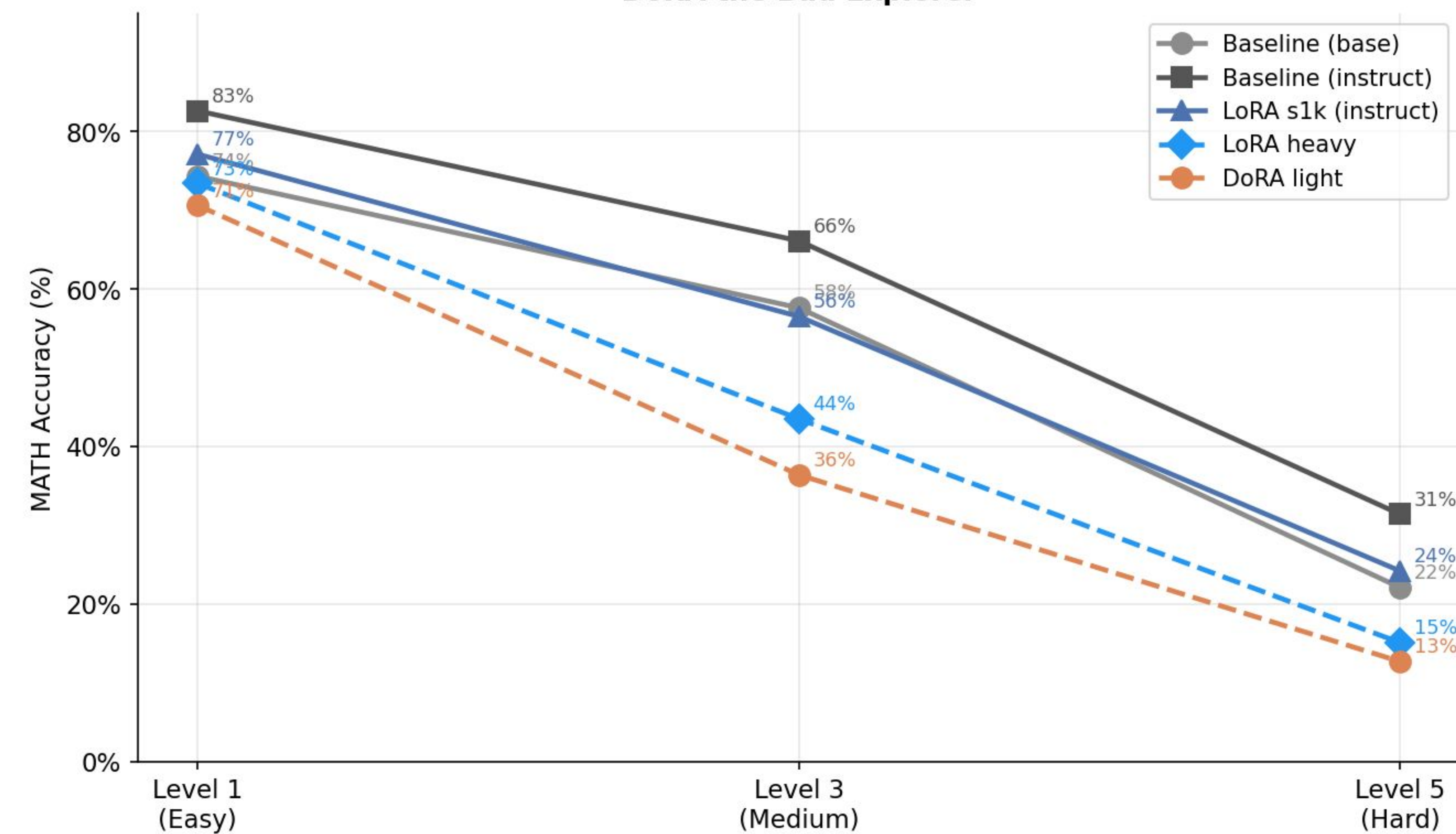
DoRA Learns to Speak with Data



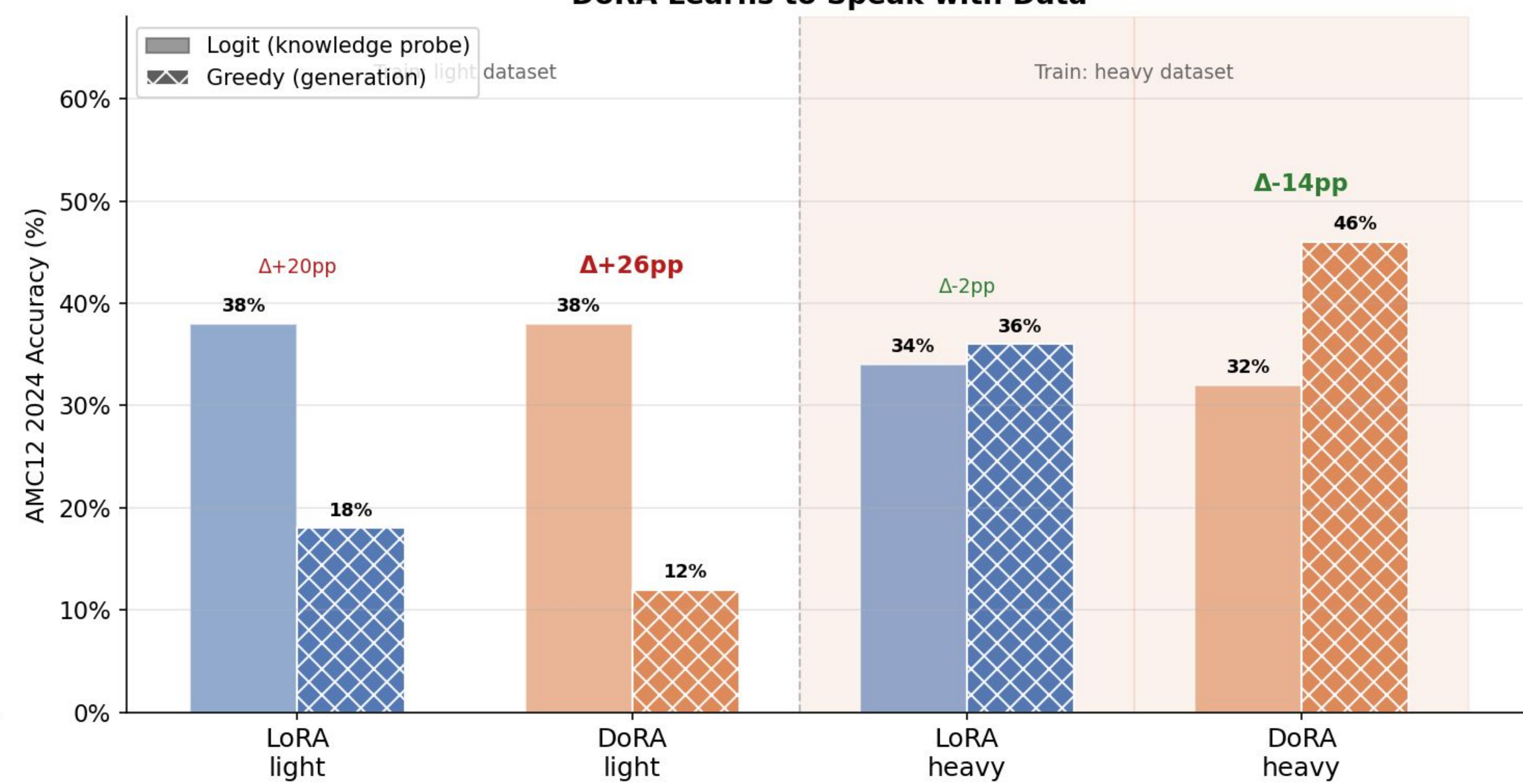
DoRA Knows, but Can't Say



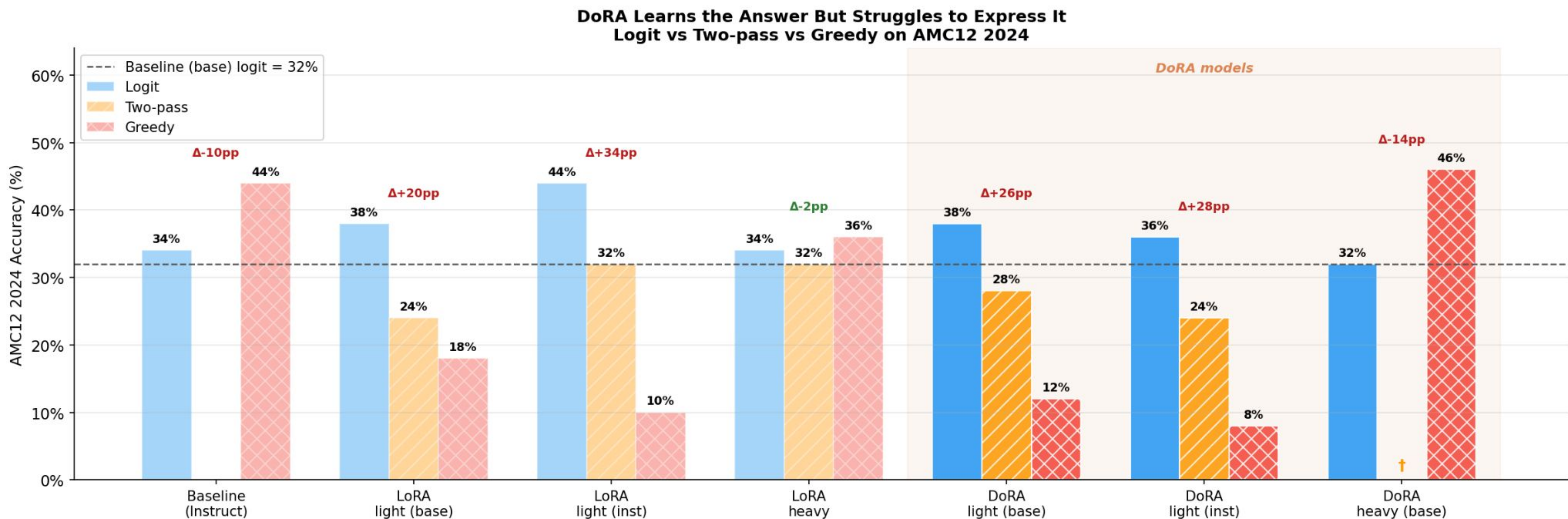
DoRA the Bad Explorer



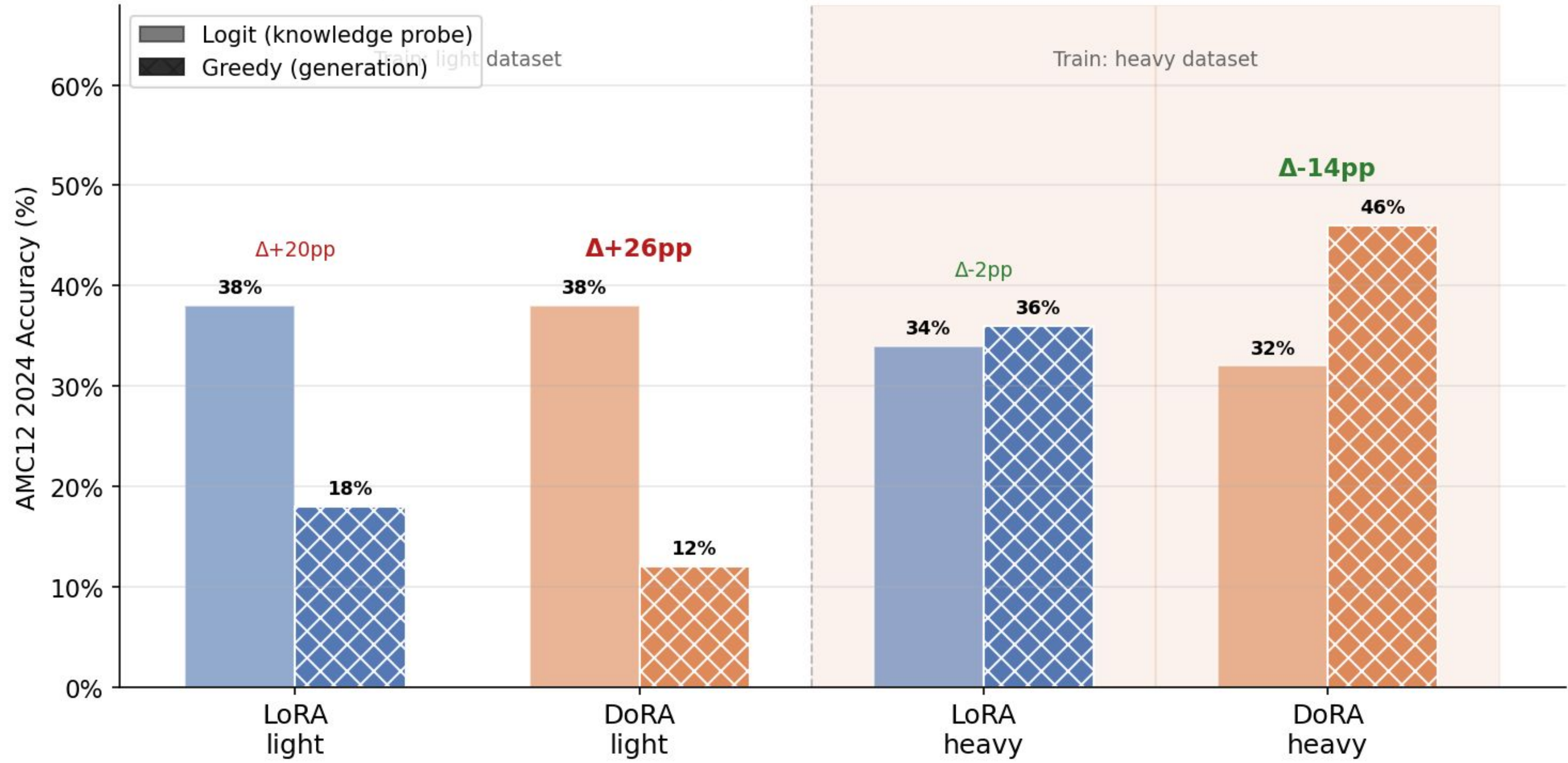
DoRA Learns to Speak with Data



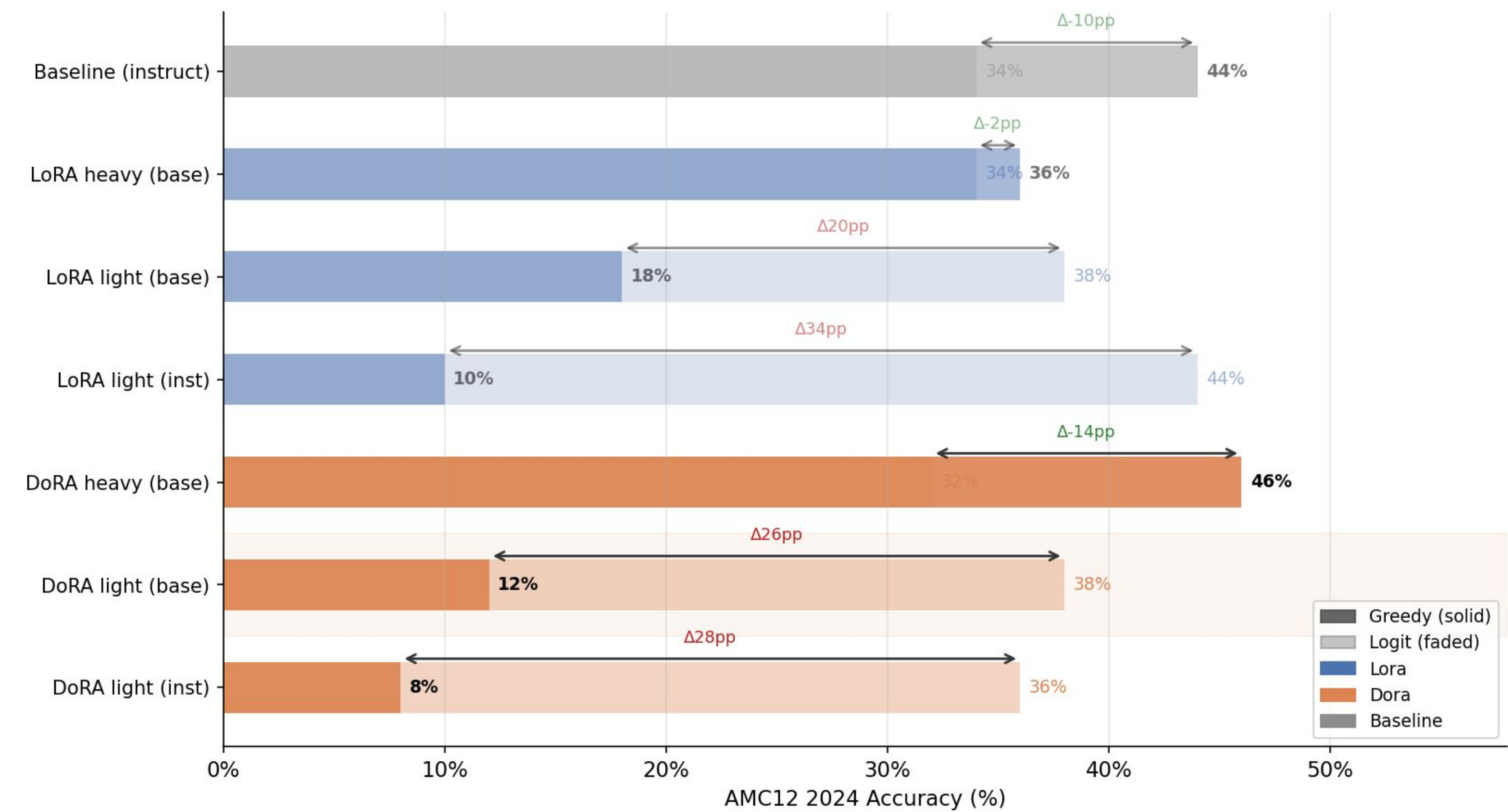
Interesting graph where for small dora/lora models a model “knows” the answer but can’t “say it”
After scaling training data, model gets better at “saying” it



DoRA vs LoRA: DoRA Knows the Answer but Can't Say It
DoRA's logit-greedy gap is larger than LoRA's — until we scale training data



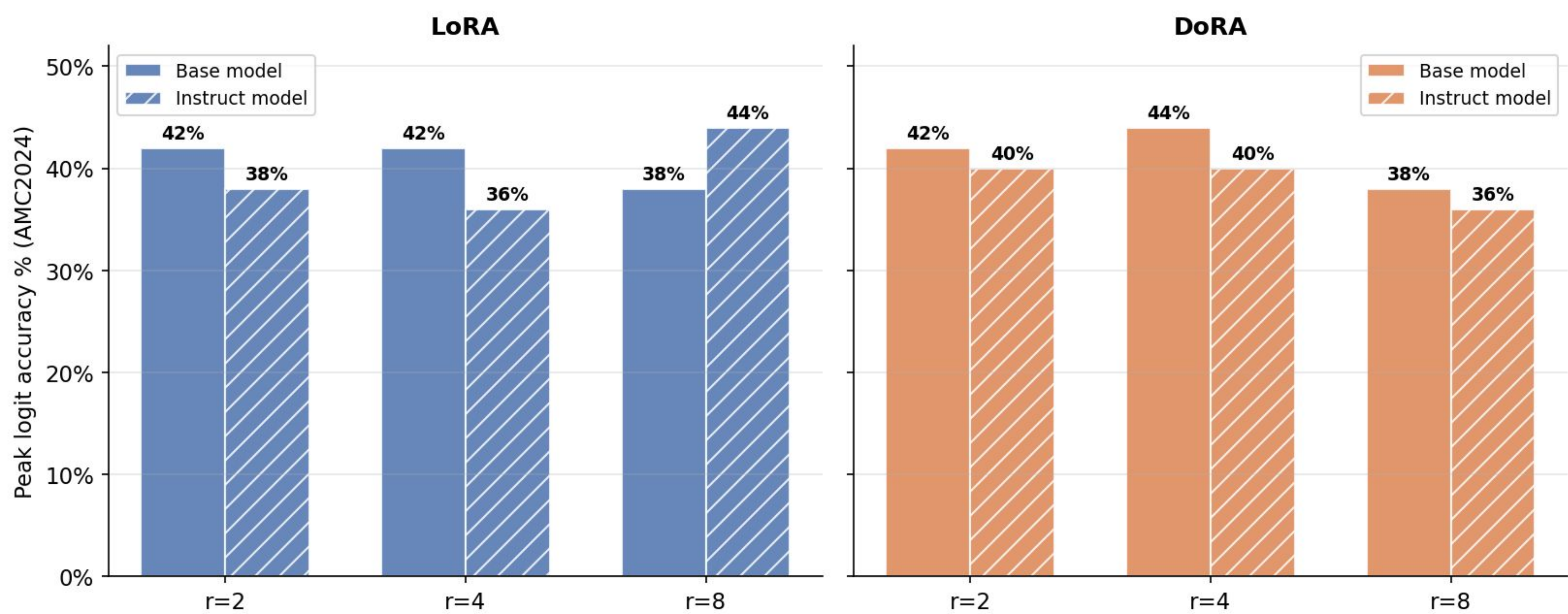
Knowledge-Expression Gap decreases with increased training data



This is a result that we see improvement

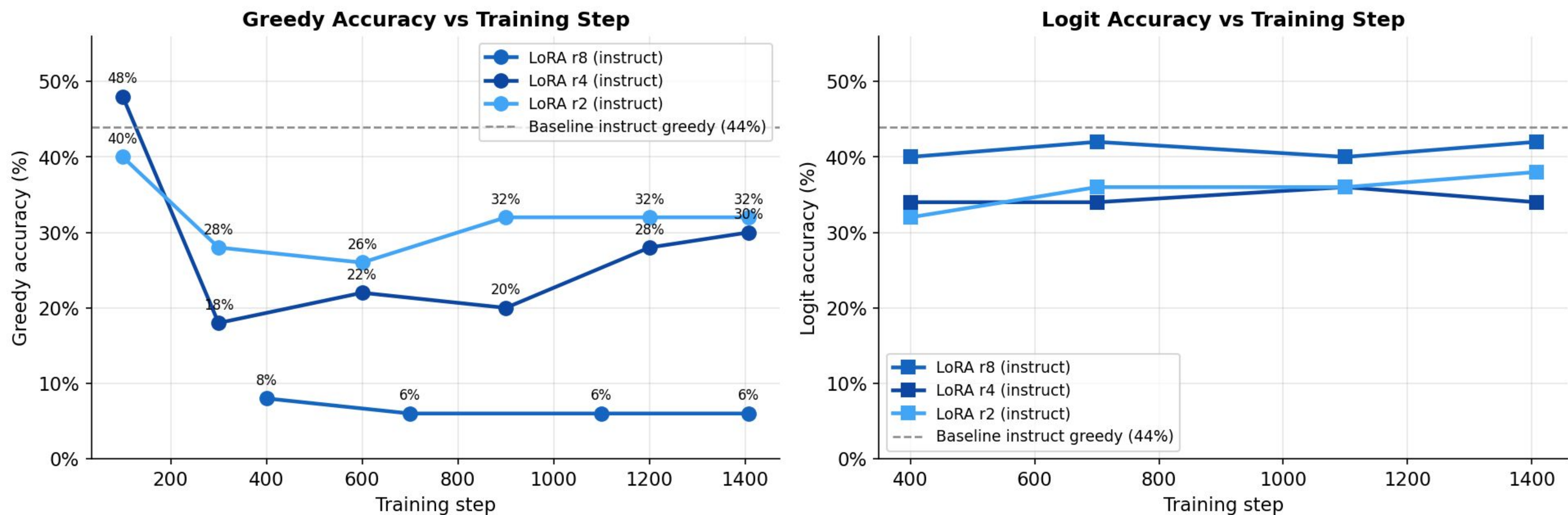
With scaling training data, the first graph encapsulates these graphs, but the first graph shows a simplified version, the second graph shows a percentage point increase from greedy to logit (from saying to knowing”)

DoRA: Logit Score is Rank-Invariant — Knowledge is Retained Regardless of Rank
 $r2 \approx r4 \approx r8$ (~42-44% logit) for DoRA; LoRA shown for comparison

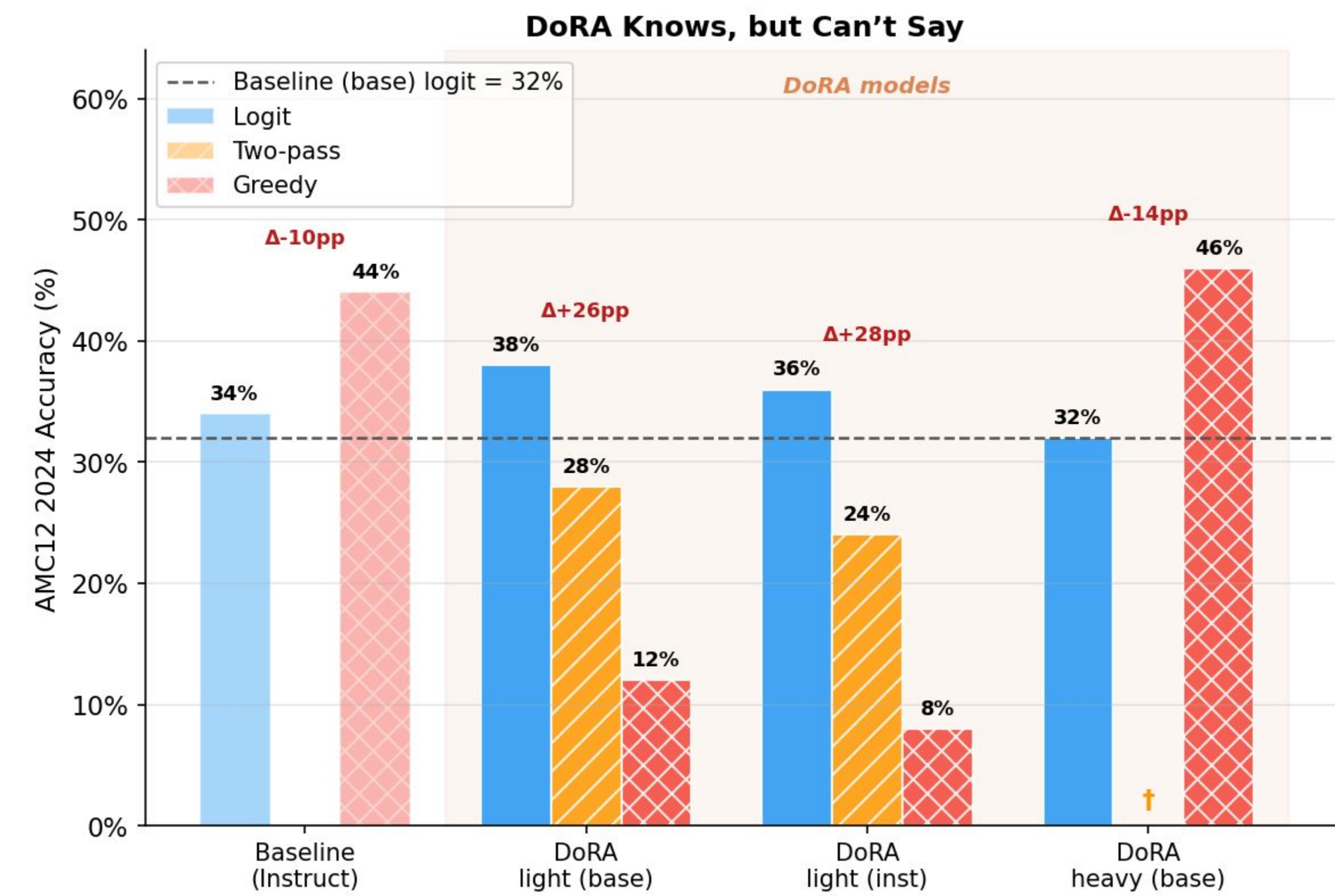
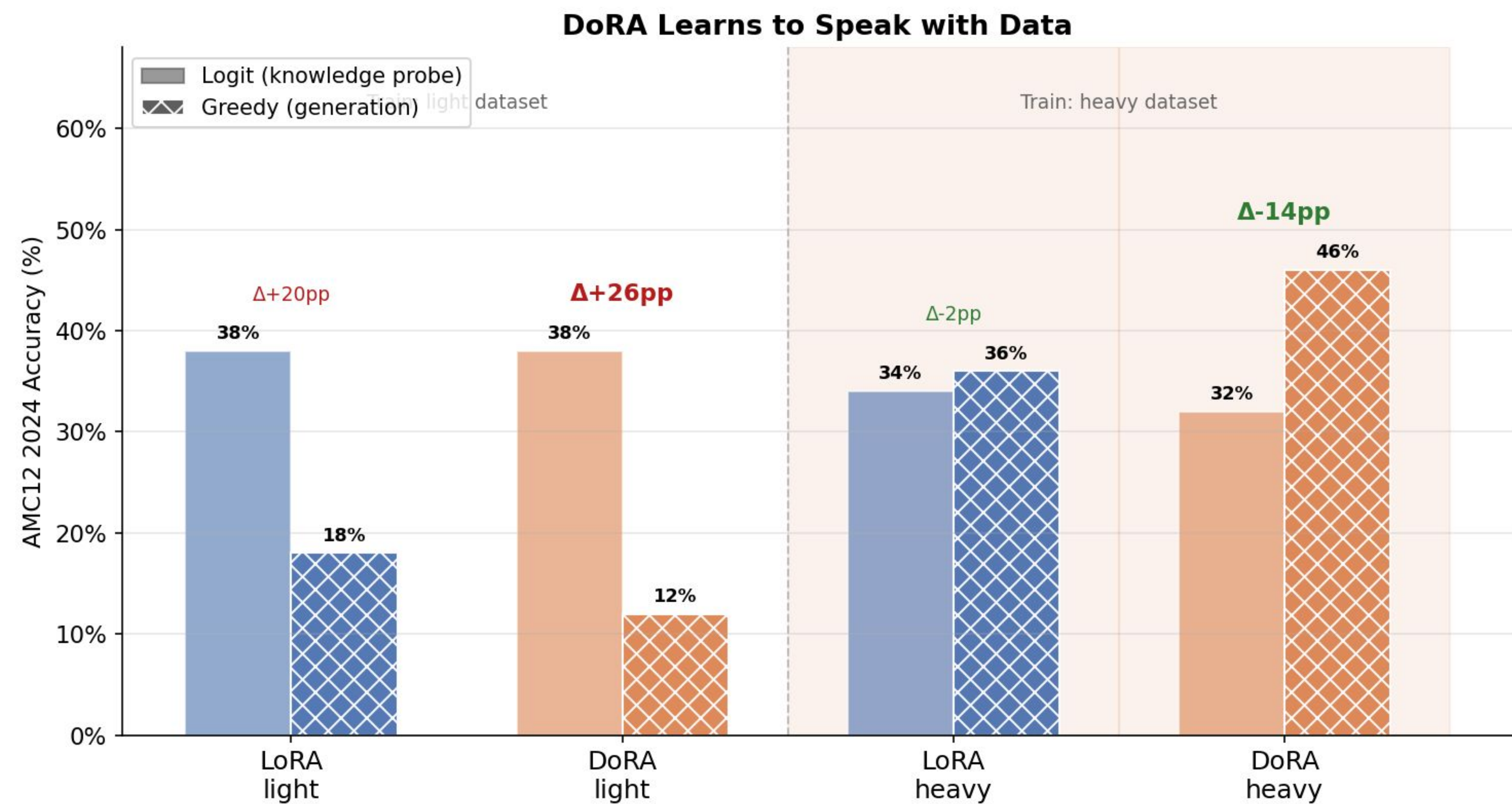
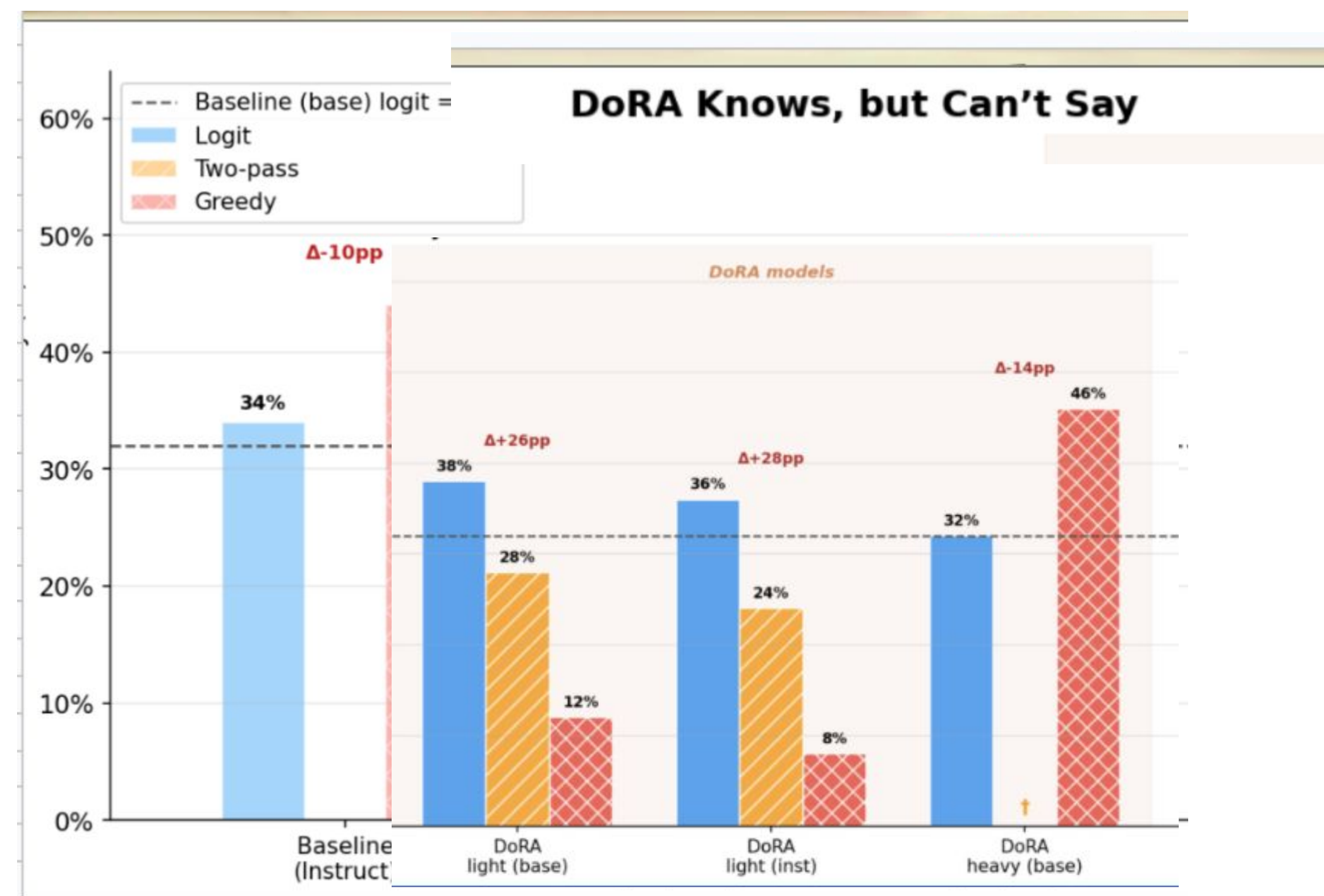


Logit score is preserved across rank. This figure isn't super necessary, but we can mention it as we talk

Fine-tuning Instruct Models: Generation Collapses While Knowledge Survives
Greedy peaks at step 100 (48%) then falls; logit stays high — same knowledge-expression gap seen in DoRA

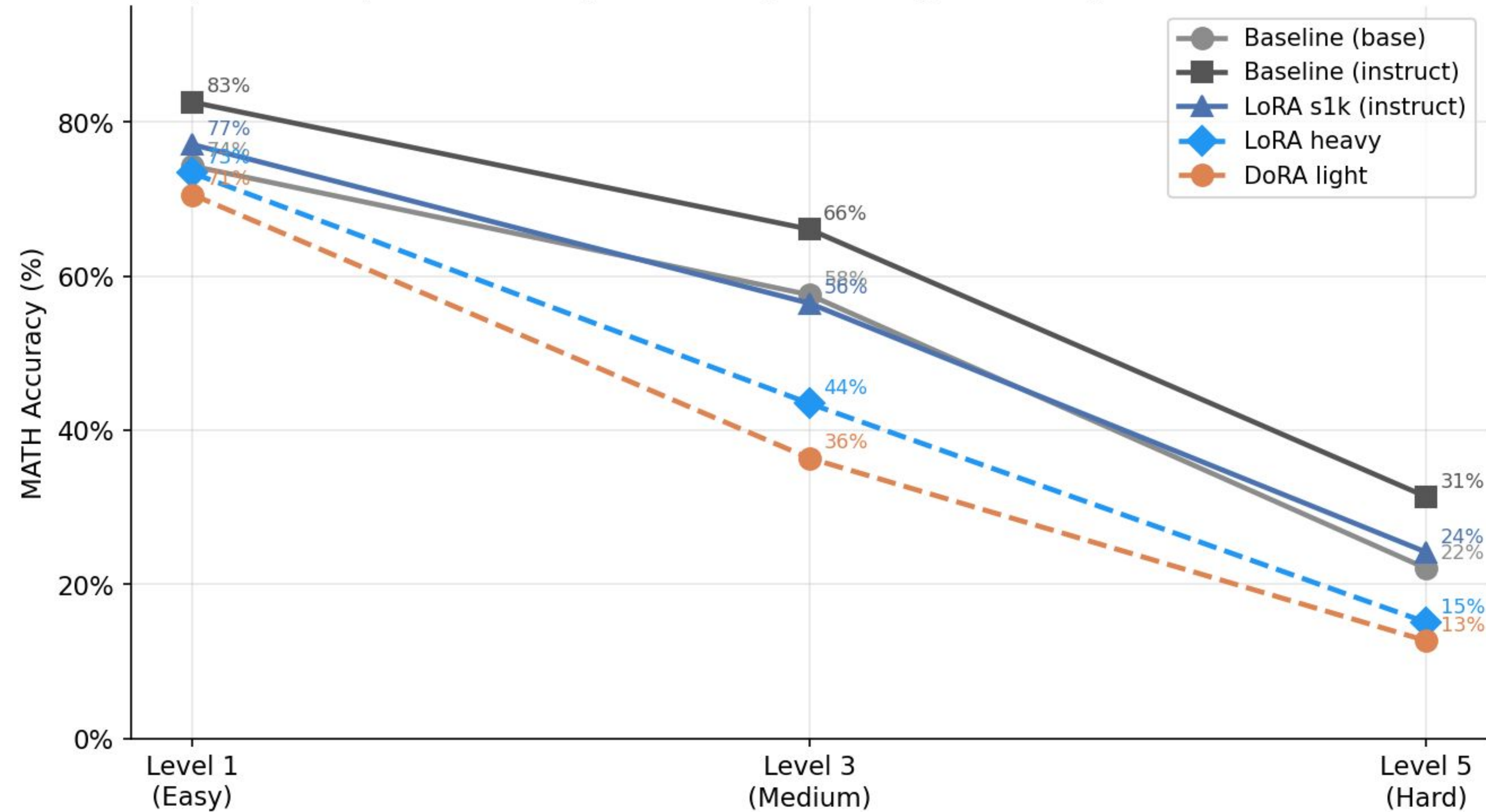


Some graph that shows that the model loses it's ability to “say”



Just add two-pass
as one bar

DoRA Loses Hard-Problem Ability Faster Than LoRA
Level-5 collapse is steepest for DoRA; LoRA s1k (small targeted data) tracks the baseline most closely



Performance degradation with hgarder problems